

ANCHORAGE INTL, AK, USA

WMO: 702730

Lat: 61.1567N

Lon: 149.9864W

Elev: 44

StdP: 100.80

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26451

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-19.4	-26.1	0.3	-19.5	-24.2	0.4	-17.6	11.0	-0.7	9.4	-1.8	1.2	10	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	22.5	15.4	20.7	14.4	19.3	13.8	16.2	21.3	15.2	19.5	14.5	18.2	3.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	10.0	17.8	13.1	9.5	16.9	12.5	9.0	16.3	45.2	21.3	42.6	19.7	40.5	18.2	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.4	7.5	-24.1	25.0	3.3	2.3	-26.5	26.7	-28.4	28.0	-30.3	29.4	-32.6	31.1		
			-24.4	17.5	3.2	1.4	-26.7	18.5	-28.6	19.4	-30.4	20.2	-32.7	21.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-7.9	-5.5	-3.2	3.2	9.0	13.3	15.3	14.2	9.8	2.7	-4.5	-6.9	(6)
(7)		DBStd	9.48	6.74	5.57	4.76	3.38	2.83	2.34	2.06	2.02	2.79	4.41	5.59	5.73	(7)
(8)		HDD10.0	2882	556	435	409	204	52	2	0	1	36	228	436	524	(8)
(9)		HDD18.3	5477	814	668	668	454	288	152	96	128	254	486	686	783	(9)
(10)		CDD10.0	451	0	0	0	0	22	101	165	132	31	1	0	0	(10)
(11)		CDD18.3	6	0	0	0	0	0	1	3	1	0	0	0	0	(11)
(12)		CDH23.3	27	0	0	0	0	0	9	14	4	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	1	2	0	0	0	0	0	(13)
(14)	Wind	WSAvg	3.0	2.6	2.7	3.1	3.1	3.6	3.5	3.1	2.8	2.8	2.8	2.8	2.7	(14)
(15)	Precipitation	PrecAvg	416	20	20	17	13	17	28	46	71	75	49	30	29	(15)
(16)		PrecMax	700	69	61	70	48	49	86	114	248	196	109	72	68	(16)
(17)		PrecMin	206	0	3	0	0	1	4	5	8	18	9	1	2	(17)
(18)		PrecStd	89	14	12	13	12	11	19	25	45	44	24	21	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.8	8.1	14.2	21.4	24.7	25.1	23.8	18.3	12.5	7.8	6.6	(19)
(20)		MCWB	3.1	2.9	4.1	7.5	11.7	15.8	17.0	16.4	12.7	8.7	4.8	3.4	3.4	(20)
(21)		2%	DB	4.2	4.2	5.8	11.6	18.5	21.7	22.7	21.3	16.4	10.7	5.7	4.4	(21)
(22)		MCWB	2.0	1.9	2.6	5.9	10.6	14.2	16.1	15.3	11.8	7.4	3.8	2.4	2.4	(22)
(23)		5%	DB	2.3	2.6	4.3	9.9	16.1	19.7	20.8	19.5	15.1	9.5	4.1	2.0	(23)
(24)		MCWB	0.7	1.0	1.5	5.0	9.4	13.3	15.0	14.4	11.1	6.7	2.5	0.6	0.6	(24)
(25)		10%	DB	0.4	1.1	2.9	8.5	14.2	18.0	19.4	18.2	13.9	8.4	2.5	0.4	(25)
(26)		MCWB	-0.7	-0.2	0.5	4.3	8.5	12.5	14.3	13.7	10.6	6.0	1.1	-0.6	-0.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.0	4.6	8.2	12.6	16.7	17.8	17.2	13.4	9.6	5.2	3.6	(27)
(28)		MDCB	5.7	4.9	7.4	13.5	20.1	23.3	23.7	22.4	16.8	11.7	6.9	5.8	5.8	(28)
(29)		2%	WB	2.1	2.1	3.0	6.4	11.0	15.0	16.4	15.9	12.4	8.0	4.0	2.3	(29)
(30)		MDCB	4.0	3.8	5.4	10.5	17.1	20.5	21.8	20.3	15.2	9.9	5.6	4.2	4.2	(30)
(31)		5%	WB	0.7	1.1	1.8	5.4	10.0	13.9	15.5	15.0	11.8	7.2	2.7	0.7	(31)
(32)		MDCB	2.0	2.4	3.9	9.3	15.3	18.6	20.0	18.7	14.2	9.1	4.0	1.7	1.7	(32)
(33)		10%	WB	-0.6	0.0	0.8	4.5	9.0	12.9	14.7	14.2	11.1	6.4	1.3	-0.6	(33)
(34)		MDCB	0.4	1.2	2.6	8.1	13.6	17.3	18.6	17.2	13.3	8.1	2.3	0.4	0.4	(34)
(35)	Mean Daily Temperature Range	MDBR	5.4	5.8	7.1	7.7	8.2	7.6	6.6	6.6	6.4	5.6	5.2	5.4	(35)	
(36)		5% DB	MCDBR	6.5	6.0	7.0	9.7	11.7	10.9	9.5	9.2	8.0	6.1	5.4	6.7	(36)
(37)		MCWBR	4.9	4.5	4.7	5.7	6.1	5.4	4.8	4.6	4.5	3.7	4.1	5.2	(37)	
(38)		5% WB	MCDBR	6.2	5.8	6.6	8.9	10.2	9.9	8.6	8.0	6.9	5.2	5.2	6.2	(38)
(39)		MCWBR	4.8	4.5	4.5	5.4	5.6	5.2	4.5	4.3	4.1	3.3	3.9	5.0	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.234	0.287	0.305	0.346	0.369	0.358	0.371	0.355	0.318	0.298	0.243	N/A	(40)	
(41)		taud	2.131	2.235	2.273	2.260	2.320	2.433	2.421	2.471	2.564	2.482	2.135	N/A	(41)	
(42)		Ebn at Noon	507	694	815	844	852	871	846	833	812	697	475	N/A	(42)	
(43)		Edh at Noon	37	65	88	109	112	102	101	89	68	52	36	N/A	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.29	0.97	2.59	4.27	5.16	5.33	4.72	3.74	2.43	1.14	0.42	0.15	(44)	
(45)		RadStd	0.04	0.14	0.28	0.38	0.48	0.35	0.43	0.41	0.34	0.14	0.07	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.78	+1.79	+1.62	N/A	N/A	N/A	-244	-302	N/A	N/A	
(47)	Regional (25 neighbors)	N/A	+2.11	+1.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page